

SELinux & AppArmor Cheat Sheet

Extra locks on what each program may touch, even as root. Find out why something was blocked, and fix it.

Which one do you have?

Distro	Uses	Quick check
RHEL, Fedora, Rocky, AlmaLinux	SELinux	getenforce prints Enforcing , Permissive or Disabled
Ubuntu, Debian, older SUSE/openSUSE	AppArmor	sudo aa-status (newer openSUSE and SLES 16 use SELinux)

SELinux: status and modes (sestatus)

```
SELinux status:                enabled
Loaded policy name:            targeted ← the normal policy: confines network services
Current mode:                  enforcing ← enforcing = blocks + logs · permissive = only logs
Mode from config file:        enforcing ← what you get after a reboot
setenforce 0                   setenforce 1 ← permissive / enforcing until reboot (for a quick test only)
/etc/selinux/config: SELINUX=enforcing ← permanent mode; after disabled, relabel: touch /.autorelabel
```

SELinux: labels (contexts)

```
$ ls -Z /var/www/html/index.html
unconfined_u:object_r:httpd_sys_content_t:s0 /var/www/html/index.html
  user      : role      : type      : level      ← the TYPE is what matters; ignore the rest
system_u:system_r:httpd_t:s0      1234 ?      00:00:02 httpd ← ps -eZ: a process of type httpd_t
```

New files take the folder's type, but **mv** keeps the old one: a file moved from **/home** stays **user_home_t**.

SELinux: finding and reading denials

```
ausearch -m avc -ts recent ← denials from the last 10 minutes (needs auditd)
journalctl -t setroubleshoot sealert -a /var/log/audit/audit.log ← plain-English explanation + fix (setroubleshoot)
type=AVC msg=audit(1791043200.412:873): avc: denied { read } for pid=2214 ← the action that was refused
comm="httpd" name="index.html" dev="dm-0" ino=135 ← the program, and the file it wanted
scontext=system_u:system_r:httpd_t:s0 ← source context: who asked
tcontext=unconfined_u:object_r:user_home_t:s0 tclass=file permissive=0 ← target's label: the wrong type here
```

SELinux: common fixes (try them in this order)

```
restorecon -Rv /var/www/html ← wrong label: reset files to the default for that path
semanage fcontext -a -t httpd_sys_content_t '/srv/www(/.*)?' ← files in a non-standard folder: record the type...
restorecon -Rv /srv/www ← ...then apply it
getsebool -a | grep httpd setsebool -P httpd_can_network_connect on ← feature off? flip a boolean (-P = keep)
semanage port -a -t http_port_t -p tcp 8081 ← service on a non-standard port
ausearch -m avc -ts recent | audit2allow -M mylocal && semodule -i mylocal.pp ← last resort
```

AppArmor (sudo aa-status)

```
apparmor module is loaded.
58 profiles are loaded.
54 profiles are in enforce mode. ← enforce = blocks + logs
  /usr/sbin/mysqld
4 profiles are in complain mode. ← complain = only logs (like SELinux permissive)
  /usr/sbin/nginx
12 processes have profiles defined. ← ...then the running processes under each mode
audit: type=1400 audit(1791043200.412:91): apparmor="DENIED" operation="open" ← a denial, from journalctl -k or dmesg
      profile="/usr/sbin/mysqld" name="/srv/mysql/ibdata1" requested_mask="r" ← which profile blocked which file

aa-complain /usr/sbin/nginx aa-enforce /usr/sbin/nginx ← log only · block again (apparmor-utils package)
apparmor_parser -r /etc/apparmor.d/usr.sbin.mysqld ← reload one profile after editing it
ls /etc/apparmor.d/ ls /etc/apparmor.d/local/ ← profiles (one per program) · your additions, kept on updates
journalctl -k | grep DENIED dmesg | grep apparmor ← find denials; aa-logprof turns them into rules
```

Remember: don't turn it off: fix the label or boolean. **setenforce 0** is for a quick test only; if the problem goes away, it was SELinux, so go and read the denial.